Louis Brouwer

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EDUCATION

Imperial College London & the Royal College of Art — **London, United Kingdom**
Innovation Design Engineering MSc & MA — Sept 2024 - Present

- Imperial College Rocketry – Technical Lead for Stringer Design on the 'Pluto' Liquid-Fueled Rocket, supervising two undergraduate engineers.
- 'Huawei Magic' project in collaboration with Huawei designing next-gen HMI products.
- Designer of 'Kinetic Canvas' an award-winning dynamic architectural lighting design that is currently being pitched for large scale architectural implementation.
- President of Imperial Surf Club. Leading and administering a student society of 10 committee and 140 members.

the University of Edinburgh — **Edinburgh, United Kingdom**
Mechanical Engineering BEng (Hons) — Sept 2020 – July 2024

- First Class Honours 1:1 – 'IMechE best student award' – awarded to highest performing graduate
- Led six-member group project team on group dissertation studying the feasibility of large-scale e-waste upcycling into robotics. 82% - Recommended for publication.
- Research on Deep Learning applied to Ground Penetrating Radar, novel results on predicting real scenarios on models trained solely on synthetic data. 82% - Recommended for publication.
- Undergraduate of the Year 2021 finalist

EXPERIENCEs

Rivan Industries — **London, United Kingdom**
Research and Development Engineer — March 2025 - Present

- Led the technical design of the carbonation system within Rivan's direct air capture (DAC) technology.
- Designed, built, and validated a 100 kW carbonation prototype achieving CO_2 capture at 5x best competitor cost , with scalability to 20x best competitor cost at >1 MW.
- Conducted techno-economic analysis to optimise cost and sizing of a lime-based DAC process, demonstrating performance and cost targets previously considered unattainable in literature.
- Managed suppliers under tight timelines and oversaw the design and manufacture of two generations of IP-grade gas–solid contactors, optimised for continuous hydroxide carbonation.

MJ Hudson (Now acquired by APEX) — **London, United Kingdom**
Summer intern – Financial Analyst — Summer 2022

- Built a new client discovery and compatibility scoring model to discover 300% more clients with 180% greater accuracy based on previously converted clients in the UK and Nordics.
- Built decks and led a 40-minute presentation for the delivery for the USA client discovery model to senior staff.

Boog Design — **Hilversum, the Netherlands**
Founder and Craftsman: — April 2015 – Sept 2022

- Over 20 unique commissions leading to realised revenues over £50,000.
- Dealing with clients through every step of the process.
- **Louisbrouwer.com**

Yacht Delivery — **North Atlantic Ocean**
Captain and First mate — May 2021 – July 2024

- Captained the 2024 delivery of a Beneteau 57 across the North Atlantic Ocean. Responsible for route planning, yacht maintenance, emergency repair, and the safety of the 5-person crew.
- Crossed the Atlantic in 2015, 2019, and 2021 As deckhand and first mate.

ADDITIONAL INFORMATION

Software	SolidWorks, TensorFlow, Python, C++, Java, Microsoft Office, Adobe creative Suite
Licenses	EASA Private Pilot's License (2018) | RYA Yachtmaster Offshore (2019)
Languages	English, Dutch (Native), Spanish (1B)
Sports	Competitive surfing - Represented Edinburgh and Imperial at BUCS 2023 and 2024 respectively
Volunteering	Cycled unsupported over 1000 miles across the whole UK to raise £33,000 for Bedales's bursary fund